Teacher answers and topic guidance

Accept any scientifically equivalent wording in spoken, signed, pointed, drawn, or written form. Every question's wording is listed beside its number below, and the I do and We do model answers match the lesson page word for word.

Starter

1. It is night here. Has the Sun switched off? Choose the best scientific prediction to test with our fixed lamp and turning globe.

Option A: The Sun stays shining all the time. Night happens at our marked place when Earth rotates on its axis so our place faces away from the Sun.

I do model answer

1. How does Earth's rotation on its tilted axis explain day and night, apparent Sun movement, time zones, length of day through the year, and seasons?

Earth rotates anticlockwise (west to east) on its axis once every 24 hours, carrying a place through sunrise (top), middle of the day (left), sunset (bottom), and night (right). Because Earth turns 360° in 24 hours (15° per hour), time zones shift by 1 hour every 15° of longitude. Because Earth's axis is tilted at 23.5°, the hemisphere tilted towards the Sun has longer days and summer, while the hemisphere tilted away has shorter days and winter.

We do model answer and checkpoint

1. Match Positions 1 to 4 in the North-Pole view to the time of day and repair the faulty claim that the Sun switches off or orbits Earth daily.

Position 1 (Top) is Sunrise (entering the sunlit left half). Position 2 (Left) is Middle of the day (facing the fixed Sun). Position 3 (Bottom) is Sunset (leaving the sunlit half). Position 4 (Right) is Night (facing away in the unlit half). Claim repair: the Sun stays shining, and night happens when Earth rotates on its axis once every 24 hours so our place faces away from the Sun.

2. In the North-Pole model with the Sun fixed on the left, Marker X is at Position 3 (bottom boundary), turning anticlockwise out of the lit half into the unlit half. Which time of day is at Marker X, and why?

Option A: Sunset, because Earth's anticlockwise rotation on its axis carries Marker X from the sunlit left half into the unlit right half.

Arrival · Supported

1. **1. Look at the arrival picture cue. Which letter marks the place on the bright sunlit half of Earth?** — Marker P is on the bright sunlit half of Earth.
2. **2. Look at Marker Q on the shaded right half of the drawing. Why is no sunlight reaching Marker Q?** — Marker Q faces away from the Sun on the unlit half of Earth.
3. **3. Think back: what do we call the imaginary pole-to-pole line that Earth spins around?** — Earth spins around its axis.
4. **4. In a classroom model, what object does the fixed lamp represent?** — The fixed lamp represents the Sun.

Arrival · Independent

1. **1. Use the arrival picture cue. Which marker is in darkness, and how does the drawing show this?** — Marker Q is in darkness because it is on the shaded right half of Earth, facing

away from the sunlight arrows.

2. **2. The Moon stays spherical all month. Why does the lit shape we see from Earth change during a month?** — The Sun always lights half of the Moon, and as the Moon orbits Earth our viewpoint shows different parts of that lit half.
3. **3. What is the difference between Earth's daily rotation and Earth's yearly orbit?** — Rotation is Earth spinning on its axis every 24 hours; orbit is Earth travelling around the Sun once a year.
4. **4. Why must we keep the model lamp still while turning the globe to investigate day and night?** — Keeping the lamp still isolates the effect of Earth's rotation so we can see how a turning planet causes day and night.

Arrival · Aspire

1. **1. Use the arrival picture cue. As Earth rotates anticlockwise in this North-Pole view, what happens as a marker at the top turns into the left half?** — A marker at the top enters the sunlit left half, modelling sunrise followed by middle of the day.
2. **2. Earth rotates 360° in 24 hours. Through how many degrees of longitude does Earth turn in 1 hour, and how does this cause time zones?** — Earth turns through 15° per hour (360° ÷ 24 = 15°), so local solar time and time zones shift by 1 hour for every 15° of longitude.
3. **3. At what angle is Earth's axis tilted, and how does this tilted axis affect the length of day through the year?** — Earth's axis is tilted at 23.5°, which makes daylight hours longer when a hemisphere tilts towards the Sun and shorter when it tilts away.
4. **4. How does Earth's tilted axis explain why the Northern and Southern Hemispheres have opposite seasons?** — When the Northern Hemisphere tilts towards the Sun it gets longer days and more direct sunlight (summer), while the Southern Hemisphere tilts away and gets shorter days (winter).

Arrival · Choice

1. **1. Look at the arrival picture cue. Point to the marker on the sunlit side of Earth.** — Marker P is on the sunlit side.
2. **2. Look at the arrival picture cue. Point to the marker on the dark shaded side of Earth.** — Marker Q is on the dark shaded side.
3. **3. Which body spins once every 24 hours to cause day and night?** — Earth spins on its axis once every 24 hours.
4. **4. Does the Sun switch off at night, or does it stay shining?** — The Sun stays shining all the time.

You do · Supported mark scheme

1. **1. Use Reference A (North-Pole view with the fixed Sun on the left). Order the four daily moments starting at Position 1: Night, Sunset, Sunrise, Middle of the day.** — 1. Sunrise (entering the light) → 2. Middle of the day (facing the Sun) → 3. Sunset (leaving the light) → 4. Night (facing away). *(Maximum 2 marks. Accept equivalent valid wording or a clear model demonstration.)*
2. **2. Complete these sentences using the word bank (day, night, rotates): When our place faces the Sun it has ___. When our place faces away it has ___. Day turns to night because Earth ___.** — When our place faces the Sun it has day. When our place faces away

it has night. Day turns to night because Earth rotates. *(Maximum 2 marks. Accept equivalent valid wording or a clear model demonstration.)*

3. **3. Explain why darkness comes each evening without the Sun switching off.** — The Sun stays shining all the time, and darkness comes when Earth rotates on its axis so our place faces away from the Sun. *(Maximum 2 marks. Accept equivalent valid wording or a clear model demonstration.)*

You do · Independent mark scheme

1. **1. Use Reference A. Track one marked place through Positions 1, 2, 3, and 4 in the North-Pole view. Name the time of day at each position and state the evidence clue.** — Position 1 (top): Sunrise, because the place enters the lit half. Position 2 (left): Middle of the day, because the place faces the Sun. Position 3 (bottom): Sunset, because the place leaves the lit half. Position 4 (right): Night, because the place faces away in the unlit half. *(Maximum 3 marks. Accept equivalent valid wording or a clear model demonstration.)*
2. **2. Explain why day turns to night using the words axis and rotation, then explain why the Sun appears to move across our sky from east to west.** — Earth spins on its axis once every 24 hours. This rotation carries each place into sunlight for day and away into shade for night. Because we rotate west to east with Earth, the Sun has an apparent movement from east to west across our sky. *(Maximum 3 marks. Accept equivalent valid wording or a clear model demonstration.)*
3. **3. Use Reference B to explain how Earth's 24-hour rotation causes time zones and why day length changes through the year.** — Earth rotates 360° in 24 hours (15° per hour), so time zones shift by 1 hour every 15° of longitude. Because Earth's axis is tilted at 23.5°, day length changes through the year as our hemisphere tilts towards or away from the Sun. *(Maximum 2 marks. Accept equivalent valid wording or a clear model demonstration.)*

You do · Aspire mark scheme

1. **1. Explain how Earth's west-to-east rotation causes both the apparent east-to-west motion of the Sun and global time zones (using 360° ÷ 24 hours = 15° per hour).** — Earth rotates west to east on its axis once every 24 hours, making the Sun appear to move east to west. Because 360° ÷ 24 hours = 15° per hour, places 15° east enter sunlight 1 hour earlier, creating 1-hour time zone steps. *(Maximum 3 marks. Accept equivalent valid wording or a clear model demonstration.)*
2. **2. Explain how Earth's 23.5° tilted axis changes the length of day through the year between the summer solstice, winter solstice, and equinoxes.** — Because Earth's axis is tilted at 23.5°, a hemisphere tilted towards the Sun spends more than half of each 24-hour turn in sunlight (longer days in summer), while the hemisphere tilted away spends less than half in sunlight (shorter days in winter). At the equinoxes, day and night are about 12 hours everywhere. *(Maximum 3 marks. Accept equivalent valid wording or a clear model demonstration.)*
3. **3. Use the tilted axis (23.5°) and Earth's orbit to explain the seasons, and state why four equal quarter turns on a non-tilted model only represent an equinox.** — Seasons happen because Earth's 23.5° tilted axis keeps the same direction in space as Earth orbits the Sun, changing both day length and how high the Sun rises. A non-tilted four-quarter model shows equal 12-hour day and night, which only happens at the equinoxes. *(Maximum 2 marks. Accept equivalent valid wording or a clear model demonstration.)*

You do · Choice mark scheme

1. **1. 1. In our North-Pole model with the Sun on the left, which half of Earth is in daylight: the left half or the right half? 2. What movement causes day and night: Earth's rotation or the Sun switching off?** — 1. Left half facing the Sun. 2. Earth's rotation on its axis. *(Maximum 2 marks. Accept equivalent valid wording or a clear model demonstration.)*
2. **2. 3. When the red marker moves from the lit half into the unlit half at the bottom (Position 3), is that Sunset or Sunrise? 4. How long does one full rotation take: 24 hours or 1 year?** — 3. Sunset. 4. 24 hours. *(Maximum 2 marks. Accept equivalent valid wording or a clear model demonstration.)*
3. **3. 5. Why does the Sun seem to cross our sky from east to west: Earth rotates west to east or the Sun orbits Earth daily? Add one short sentence using the word apparent.** — Earth rotates west to east. The Sun's daily path across the sky is an apparent movement caused by Earth spinning. *(Maximum 2 marks. Accept equivalent valid wording or a clear model demonstration.)*

Exit ticket · Supported

1. **1. In the North-Pole model, Place X is on the unlit right half of Earth. Is Place X in day or night, and what does Earth do so Place X reaches daylight?** — Place X is in night. Earth rotates on its axis, carrying Place X around into the sunlit half.
2. **2. How many hours does one full rotation of Earth take?** — One full rotation of Earth takes about 24 hours.
3. **3. Why does the Sun not need to switch off for us to have night?** — The Sun stays shining all the time, and we have night when our place on Earth rotates away from the Sun.

Exit ticket · Independent

1. **1. Explain why Place X on the unlit side of Earth has night and how it reaches sunrise and middle of the day.** — Place X faces away from the Sun so no direct sunlight reaches it. As Earth rotates anticlockwise (west to east), Place X turns into the lit half at sunrise and faces the Sun at middle of the day.
2. **2. Why does the Sun appear to move across our sky from east to west each day?** — We rotate from west to east with Earth, so our viewing direction sweeps past the Sun and makes it appear to move from east to west.
3. **3. How does Earth's 23.5° tilted axis change the length of day through the year?** — When our hemisphere tilts towards the Sun we stay in the lit half longer (longer summer days), and when it tilts away we stay in the lit half for less time (shorter winter days).

Exit ticket · Aspire

1. **1. Explain Place X's night and later daylight using both the space viewpoint (rotation) and the Earth observer's viewpoint (apparent movement).** — From space, Earth rotates west to east on its axis, carrying Place X from the unlit half into sunlight. From Place X, the observer turns with Earth and sees the Sun appear to rise in the east and cross the sky to the west.
2. **2. How does Earth's 360° rotation in 24 hours (15° per hour) explain time zones?** — Because Earth rotates 15° of longitude every hour, places 15° further east turn into sunlight 1 hour earlier, so global time zones step by 1 hour every 15°.

3. **3. How does Earth's tilted axis (23.5°) explain changing length of day through the year and the seasons?** — Earth's 23.5° tilted axis means that as Earth orbits the Sun, the hemisphere tilted towards the Sun gets longer daylight hours and more direct sunlight (summer), while the hemisphere tilted away gets shorter daylight hours and lower sunlight (winter).

Exit ticket · Choice

1. **1. Place X is on the shaded half facing away from the Sun. Choose: is Place X in day or in night?** — Place X is in night.
2. **2. What carries Place X from night into sunrise? Choose: Earth rotates on its axis / The Sun switches back on.** — Earth rotates on its axis.
3. **3. How long is one full rotation of Earth? Choose: 24 hours / 365 days.** — 24 hours.

Exit pupil voice

1. What helped you today, and what would you change? — Accept any pointed, spoken, signed, drawn, written, or chosen reflection on what helped and what could change next time. A pupil may pass.

Topic guidance and Key Stage 3 progression

- **Building on primary knowledge to reach Key Stage 3:** At primary level, pupils learn that Earth spins once every 24 hours to cause day and night. At Key Stage 3, connect this spin to **time zones** (360° ÷ 24 hours = 15° of longitude per hour) and connect Earth's **23.5° tilted axis** to changing **length of day through the year** and the **seasons**.
- **Does the Sun move around Earth each day?** Keep the model Sun fixed on the left and show that Earth rotates west to east on its axis once every 24 hours, creating the Sun's apparent east-to-west motion across the sky.
- **Tilted axis, day length, and seasons:** When the Northern Hemisphere tilts towards the Sun at the summer solstice, more than half of each Northern latitude circle is in sunlight (longer days and summer) while the Southern Hemisphere tilts away (shorter days and winter). At the equinoxes, day and night are about 12 hours worldwide.

Made by Matt